

# When is local learning enough?

## Stability and spatial error bounds for conditional sampling on graphs

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A field may contain a million coordinates while the question concerns one. How much must be read, and how accurately must local laws be learned? I seek to recruit a postdoctoral mathematician from outside Sweden to answer this for conditional sampling on graphs. Separately learned local scores (gradients of conditional log-densities) may be incompatible: they need not come from one joint density. Accuracy where the data lie need not prevent a sampler from visiting regions where its drift is wrong.

We ask: *when does local learning suffice for accurate conditional sampling?* Mathematically, the project develops a perturbation theory for Gibbs samplers with incompatible local conditionals. Two aims drive it: **(a)** turn score error measured under the true law into a guarantee for the sampler actually run, with nonconvex, incompatible updates and an artificial boundary where the neighbourhood is cut; **(b)** choose radius, step size and run length from the desired accuracy. Preliminary results verify the feedback mechanism exactly in a two-site Gaussian model with incompatible drifts and numerically estimate its constants for a nonconvex family. In a Gaussian chain our truncation bound decays at the optimal rate, whereas a cruder estimate needs an asymptotically 19 times larger radius at neighbour correlation 0.9, and unboundedly larger as correlations lengthen.

### 1. Research environment at Lund University

**1.1 Host and intellectual leadership.** I formulated this project to connect two strands of my research: rigorous error analysis of interacting stochastic systems and the development and analysis of learned dynamical models. With Katsoulakis and Plecháč, I established second-order weak error estimates for coarse-grained lattice dynamics, governed by coarse-graining scale relative to interaction range [1]. My work on stochastic closures [2] and interacting traffic models [3] also connects local interactions with macroscopic behaviour, and I independently developed lattice-free stochastic dynamics [4]. With my doctoral student Donglin Liu, I developed GS-SINDy, combining Earth Mover's distance and group similarity with convergence analysis [5]; our work with Oskar Åström develops graph-enhanced diffusion models for spatiotemporal imputation [6]. The connecting question is how errors in local approximations propagate through an interacting system; here those approximations are learned and may be inconsistent.

I carried out the preliminary Gaussian analysis, nonconvex numerical investigations and parameter example below. I will lead the mathematical programme, working with the fellow on boundary transfer (Theorem A), defect estimates, certification and the synthesis of Theorems A and B. The fellow will lead the nonconvex defect analysis, supported by Friesen's coupling expertise and Katsoulakis's specialist input on score-based models.

**1.2 Group, collaboration and supervision.** I am Senior Lecturer and Docent in Mathematics at Lund's Centre for Mathematical Sciences (since 2011). My supervision includes Donglin Liu (main supervisor, PhD 2025), Oskar Åström (main supervisor, 2023–), four further doctoral students as co-supervisor, and three postdocs as supervisor or co-supervisor, including Shuzhen Li (main supervisor, 2025–). The fellow will join this active environment for learned dynamics and graph modelling, meeting me weekly. Lund has agreed reduced teaching duties for my externally funded research.

Martin Friesen, Senior Lecturer in Mathematical Statistics, has agreed to be second mentor, contributing expertise in ergodicity, concentration and interacting particle systems to the nonconvex coupling analysis. Markos A. Katsoulakis, Professor and Director of the Center for Applied Mathematics and Computation at UMass Amherst, has confirmed participation as co-mentor. We have eight joint papers; his recent work links score-based models to Wasserstein proximal operators

[7]. Both co-mentors will join monthly meetings. Katsoulakis will host a four-week visit in year 1 and visit Lund in year 2.

As WASP Lighthouse Manager (2026–2029) and coordinator of the LTH Numerical Analysis and Lund eScience Hub seminars, I connect the fellow to wider mathematical and computational networks. LUNARC supports certification computations and COMPUTE provides a scientific-computing research school. Pablo Villanueva-Perez (Lund, ERC Starting Grant holder) has confirmed an optional MAX IV study linked to sparse X-ray reconstruction [8].

## 2. Project and research area

**2.1 Objective and precise gap.** For a fixed target  $S$ , using only its radius- $r$  neighbourhood, we seek  $W_1$ -error of order  $\varepsilon + \sqrt{h} + (1+r)^p e^{-\kappa r}$  plus a finite-time term. Here  $\varepsilon$  measures score error under the true conditional (reference) law,  $h$  is the Euler step size, and  $\kappa$  is the effective spatial decay rate. We target spatially mixing finite-range fields with nonconvex one-site potentials, under explicit residual derivative control. The result should specify admissible radius, step size and run length for a desired accuracy.

Dobrushin comparison [9, 10], sparse block samplers [11], Wasserstein perturbation theory [12] and reflection coupling [13, 14] provide starting tools. Incompatible conditionals also occur in iterative imputation [15]. Cui–Liu–Tong [16] already bound marginal errors by target-law score discrepancies when the approximating density has suitable locality; their result compares scores of joint densities. Our learned drifts need not be scores of their sampler’s invariant law. Stable localised samplers [17] and radius–statistical-error tradeoffs for reverse diffusion [18] address related questions. Xi [19] assumes weighted drift error along the learned process; we seek to derive such control from reference-law errors and residual regularity. Population score loss alone cannot ensure Langevin accuracy [20], and score accuracy does not certify numerical stability of diffusion sampling [21].

**The proposed advance** is a finite-time local marginal guarantee for the implemented random-scan Euler sampler, allowing incompatible, nonconvex block drifts and an artificial boundary, uniformly in graph size. Standard comparison supplies the initial boundary estimate. The research challenge is to certify defect envelopes with feedback below the contraction margin, retaining explicit decay through both field and sampler resolvents. Useful parameter ranges and full accuracy multipliers, rather than the perturbation recursion alone, are the central targets. Verifiable block conditions should give an abstract theorem, and the sinusoidal family of Section 2.4 its first certified nonconvex example.

**2.2 Sampler, assumptions and baseline.** Let  $X$  have a smooth, positive, finite-range Gibbs density on a bounded-degree graph with  $N$  coordinates. Given  $X_O = y$ , write  $\pi_y = \mathcal{L}(X_U \mid X_O = y)$  and partition  $U$  into bounded blocks  $B_i$ . The target  $S$  is a fixed union of blocks, also identified with their indices. Retain its radius- $r$  neighbourhood  $I = I_r(S)$  in the block dependency graph and clamp other unobserved coordinates to  $b$ , obtaining  $\pi_{y,r}^b$ . Observations remain fixed. Each update chooses one of the  $k_r = |I|$  blocks uniformly, freezes its local boundary  $z$ , and takes  $n$  steps with learned drift  $s_i$ :

$$Z_{\ell+1} = Z_\ell + h s_i(Z_\ell, z, y) + \sqrt{2h} \xi_{\ell+1}, \quad \xi_\ell \sim N(0, I_{|B_i|}), \quad \tau = nh. \quad (1)$$

Let  $\mu_T$  be the law after  $T$  updates. On  $S$ ,  $W_1$  uses the sum of block Euclidean distances. With  $s_j^*$  the true conditional score at the appropriate boundary, the inputs are

$$\varepsilon_{y,j}^2 = \mathbb{E}_{\pi_y} \|s_j - s_j^*\|^2, \quad \varepsilon_{y,j}^b = \mathbb{E}_{\pi_{y,r}^b} \|s_j - s_j^*\|.$$

These losses concern the specified  $y$ ; unconditional training loss does not ensure accuracy for rare observations. Finite-sample generalisation is out of scope.

For a nonnegative matrix  $A$ , set  $\|A\|_w = \sup_i w_i^{-1} \sum_j A_{ij} w_j$ , with  $0 < w_- \leq w_i \leq w_+$  uniformly in  $N$ . Let  $C$  bound blockwise  $W_1$  influences of the conditional specification, with  $\|C\|_w \leq q < 1$ . Assume uniform block dimensions, interaction range, local Lipschitz and linear-growth constants of true scores, and fourth moments per block under full and clamped reference laws. Also require

$\sup_i \mathbb{E} \|\widehat{X}_{0,i}\|^4 \leq K_0$ . Bounds are uniform over the bounded observations and clamping values considered; discretisation constants may depend on  $K_0$ .

**Baseline (months 1–4).** If learned drifts are  $L$ -Lipschitz and  $\widetilde{m}$ -contractive in their own block, with cross-block Lipschitz matrix  $a$ , synchronous comparison gives the intended estimate

$$W_1((\mu_T)_S, (\pi_{y,r}^b)_S) \leq \frac{h}{1 - \rho_h} \sum_{i \in S} [(I - A_h)^{-1}(\varepsilon_y^b + c\sqrt{h}\mathbf{1})]_i + K_S M_0 e^{-\gamma_h T/k_r}. \quad (2)$$

Here  $\rho_h = (1 - 2\widetilde{m}h + L^2 h^2)^{1/2} < 1$  for  $0 < h < 2\widetilde{m}/L^2$ ,  $A_h = ha/(1 - \rho_h)$ ,  $\|A_h\|_w \leq 1 - \zeta$ , and  $\gamma_h = (1 - \rho_h^n)(1 - \|A_h\|_w)$ . For a coupling with  $X_0 \sim \pi_{y,r}^b$ , set  $M_0 = \max_{i \in I} w_i^{-1} \mathbb{E} \|\widehat{X}_{0,i} - X_{0,i}\|$ . The  $O(\sqrt{h})$  term uses Lipschitz regularity and moment bounds; an  $O(h)$  refinement would require stronger assumptions. The baseline yields a unique invariant law and tests the error decomposition before learned-drift contractivity is removed.

Taking the learned kernel as the contractive reference in perturbation theory [12] allows averaging defects under the exact stationary law. Exact-kernel contraction instead places defects along the learned trajectory. Nonconvexity alone excludes neither approach; Theorem B seeks controlled feedback when direct learned-kernel contraction is unavailable or gives inferior constants.

**2.3 Theorem A: spatial transfer across an artificial boundary.** Set  $D_I = (I - C_{II})^{-1}$ ,  $m_j^b = \mathbb{E}_{\pi_y} \|X_{B_j} - b_j\|$  for  $j \notin I$ ,  $t_I = D_I C_{I,I^c} m_{I^c}^b$  and  $t_{I^c} = m_{I^c}^b$ . Standard Gibbs comparison [10], extending the clamped law by  $b$ , bounds observable discrepancies by  $\sum_k \text{Lip}_k(\varphi) t_k$ . Applied to  $\|s_j - s_j^*\|$  with block Lipschitz constants  $L_{jk}$ , followed by Cauchy–Schwarz, it gives  $\varepsilon_{y,j}^b \leq \varepsilon_{y,j} + \sum_k L_{jk} t_k$ . Thus for the nonnegative sampling resolvent  $R$  in the baseline or Theorem B,

$$\sum_{i \in S} [R\varepsilon_y^b]_i \leq \sum_{i \in S} [R\varepsilon_y]_i + \sum_{i \in S} [RLt]_i, \quad W_1((\pi_{y,r}^b)_S, (\pi_y)_S) \leq \sum_{i \in S} t_i. \quad (3)$$

**Target.** Derive exponential kernel bounds for  $D_I$  and  $R$  from verifiable block conditions and, for polynomial neighbourhood growth and uniformly bounded finite-range  $L$ , prove

$$\sum_{i \in S} [RLt]_i \leq K_S M_b (1 + r)^p e^{-\kappa r}, \quad M_b = \sup_{j \notin I} m_j^b, \quad \kappa = \min\{\kappa_D, \kappa_R\}. \quad (4)$$

Constants and  $p$  must be explicit and uniform in  $N, r$ . On general bounded-degree graphs, establish the corresponding weighted tail bound. Keeping the two resolvents separate distinguishes field correlation from sampler-error propagation. We will quantify decay-rate deterioration under residual feedback and identify conditions retaining the reference rate.

**2.4 Theorem B: nonconvex, incompatible learned updates.** A drift can agree with  $-u$  on  $[-R_0, R_0]$  and point outward beyond  $2R_0$  while its Gaussian-weighted error tends to zero. We therefore start with a confined, explicitly structured family:

$$V(x) = \sum_i [\alpha x_i^2/2 + \eta(1 - \cos x_i)] - \sum_{\{i,j\} \in E} J_{ij} x_i x_j, \quad \eta > \alpha > 0, \quad \sup_i \sum_j |J_{ij}| < \alpha. \quad (5)$$

Here  $J$  is symmetric and finite-range. One-site potentials are nonconvex near  $\pm\pi$ , but tails are quadratically confined; a stronger mixing condition is also required. For scalar blocks, write  $s_i(u, z) = -\alpha u + \sum_j J_{ij} z_j + g_i(u, z)$  and  $g_i^*(u) = -\eta \sin u$ . For fixed  $B > \eta$ , let  $\mathcal{G}$  consist of local  $C^1$  functions with  $|g_i| \leq B$ ,  $\|\partial_u(g_i - g_i^*)\|_\infty \leq \lambda_1$  and  $\|\partial_{z_j} g_i\|_\infty \leq \lambda_2$  for each neighbour. Degree factors are explicit. If  $\lambda_1 < \eta - \alpha$ , every such drift fails global one-site Euclidean contractivity. Explicit residuals with own-coordinate dependence are

$$r_i(u, z) = g_i - g_i^* = a_i + c_i \tanh u + \sum_j b_{ij} \tanh z_j, \quad |a_i| + |c_i| + \sum_j |b_{ij}| \leq B - \eta,$$

with  $|c_i| \leq \lambda_1$  and  $|b_{ij}| \leq \lambda_2$ . Nonzero cross coefficients give incompatible drifts: the difference of cross derivatives is  $b_{ij} \text{sech}^2 x_j - b_{ji} \text{sech}^2 x_i \not\equiv 0$ . We will certify examples with both own-coordinate and cross-block residual dependence.

Derivative accuracy relative to the true score is a global assumption beyond population loss. Bounded architectures can enforce amplitude and cross-derivative bounds, but cannot by themselves certify that accuracy against an unknown score. The known-score family is a proof and validation

model; certificates will report admissible derivative tolerances, distinguishing assumptions from training-observable quantities. Broader potentials will be pursued through the block criterion, without assuming arbitrary multimodality tractable.

**Defect mechanism.** The exact one-block diffusion  $P_i$  over time  $\tau$  preserves  $\pi_{y,r}^b$  when the joint starting law is  $\pi_{y,r}^b$ . In a common metric  $\mathfrak{d}$  satisfying  $c_-|u-v| \leq \mathfrak{d}(u,v) \leq c_+|u-v|$ , seek initial-state contraction  $\theta_i \leq \theta_* < 1$  and boundary influences  $M_{ij}$  ( $M_{ii} = 0$ ). Couple learned and exact updates from one state, then exact updates from different states, using reflection coupling where needed. If the one-update defect obeys  $b_i \leq \tilde{b}_i + e_i$  and  $\tilde{b}_i$  has blockwise  $\mathfrak{d}$ -Lipschitz constants  $\Lambda_{ij}$ , coupling to the stationary reference gives

$$\mathbb{E}b_i(\widehat{X}) \leq \bar{b}_i + \sum_j \Lambda_{ij}d_j + c_2\sqrt{h}, \quad \bar{b}_i = \mathbb{E}_{\pi_{y,r}^b}\tilde{b}_i, \quad d_j = \mathbb{E}\mathfrak{d}(\widehat{X}_j, X_j).$$

Thus the defect is averaged under the reference law at the price of a feedback term  $\Lambda d$ ; Theorem B must keep  $M + \Lambda$  within the contraction margin. For the scalar family a Euclidean candidate envelope is

$$\tilde{b}_i(x) = \int_0^\tau e^{\ell(\tau-t)} \mathbb{E}_x|r_i(Y_t)| dt, \quad \ell = \eta - \alpha + \lambda_1, \quad (6)$$

where  $Y$  is the exact block diffusion with frozen boundary. Stationarity gives  $\bar{b}_i \leq (e^{\ell\tau} - 1)\varepsilon_{y,i}^b/\ell$ . Flow sensitivity should yield  $\Lambda_{ii} \leq C(\tau)\lambda_1$  and  $\Lambda_{ij} \leq C(\tau)(\lambda_2 + \lambda_1|J_{ij}|)$ , with metric factors included when using reflection distances. A constant residual has a constant envelope even when its exact defect varies with the starting point. Confinement, bounded  $g_i$  and the initial fourth-moment bound will control the Euler defect uniformly over the transient and graph size.

**Target.** Fix  $\alpha, \eta, B$ , degree and an admissible  $\tau$ . Prove explicit positive thresholds for  $h$ ,  $\sup_i \sum_j |J_{ij}|$ ,  $\lambda_1, \lambda_2$  such that, uniformly over  $\mathcal{G}$ ,

$$H = I - \text{diag}(\theta_i), \quad F = H^{-1}(M + \Lambda), \quad \|F\|_w \leq 1 - \zeta, \quad H^{-1}\bar{b} \leq c_1\varepsilon_y^b.$$

The random-scan recursion should then imply

$$W_1((\mu_T)_S, (\pi_{y,r}^b)_S) \leq c_-^{-1} \sum_{i \in S} \left[ (I - F)^{-1} (c_1\varepsilon_y^b + c'_2\sqrt{h}\mathbf{1}) \right]_i + K_S M_0 e^{-\gamma T/k_r}, \quad (7)$$

where  $M_0$  is defined in  $\mathfrak{d}$  and  $\gamma = \min_i \{1 - \theta_i - w_i^{-1} \sum_j (M_{ij} + \Lambda_{ij})w_j\} \geq \zeta(1 - \theta_*) > 0$ . Constants depend on structural bounds,  $K_0$  and target size, not  $N, r$ . This controls target error; mixing of the learned chain needs a separate argument. We will compare certified parameter ranges, multipliers and decay rates with direct reflection coupling of the learned kernel.

**Preliminary evidence and what it does not prove.** For two Gaussian sites with scores  $-x_i + cx_j$ , perturb the cross coefficients to  $c + \delta$  and  $c - \delta$ , where  $0 \leq c < 1$  and  $0 < |\delta| < 1 - c$ . Exact inner integration gives  $\theta = e^{-\tau}$ , off-diagonal entries  $c$  in  $H^{-1}M$  and  $|\delta|$  in  $H^{-1}\Lambda$ , and stationary coupled discrepancy  $d_i \leq \sqrt{2/\pi} \varepsilon / (1 - c - |\delta|)$ , with  $\varepsilon = |\delta|/\sqrt{1 - c^2}$ . This verifies incompatible-drift feedback for every  $\tau > 0$ . For (5) with  $\alpha = 1$ ,  $\eta = 1.5$ ,  $\tau = 4$ , my generator discretisation on  $[-14, 14]$  (561 points) estimated  $\theta \approx 0.39$  and boundary-field sensitivity  $\approx 2.01$ , searching  $|x| \leq 10$  and 16 fields. Refinement to 801 points and 24 fields was stable; the code recovered Gaussian constants at  $\eta = 0$ . Quadrature gives a separate equilibrium criterion  $\sum_j |J_{ij}| < 1/\sup_f \text{Var}_{\pi_f} U \approx 0.393$ . These are numerical estimates, not global certificates. On a chain with nearest-neighbour  $J_{ij} = 0.05$  and  $\lambda_1 = \lambda_2 = 0.005$ , preliminary constants give  $\|F\|_w \approx 0.81$ ,  $c_1 \approx 21$  and score multiplier about 110: admissibility is not yet practical tightness. We will optimise  $\tau$  and envelopes, using analytic tails, validated discretisation and bounds between grid points. Field reduction must respect  $(u, f) \mapsto (u + 2\pi, f + 2\pi\alpha)$ . Reflection distances are an alternative when short-time Euclidean bounds exceed one.

**2.5 Sharp truncation benchmark and computational consequence.** For  $X_{j+1} = aX_j + \sqrt{1 - a^2}\xi_{j+1}$  in stationarity,  $0 < a < 1$ , observing  $X_d = \pm 1$  gives  $\pi_\pm = N(\pm a^d, 1 - a^{2d})$  at the origin. A sampler unable to read  $X_d$  has the same output  $\nu$  in both cases, so  $\max\{W_1(\nu, \pi_+), W_1(\nu, \pi_-)\} \geq a^d$ . For Gaussian precision  $Q$  with nonpositive off-diagonals,  $C = I - D_Q^{-1}Q$  and  $(I - C)^{-1} = \Sigma D_Q$ ,

where  $D_Q = \text{diag}(Q)$  [22]. On the infinite chain the weights are  $(1 + a^2)a^{|i-j|}/(1 - a^2)$ , matching the lower exponent. The row-sum estimate  $q^r$ ,  $q = 2a/(1 + a^2)$ , requires an asymptotically larger radius by  $\log(1/a)/\log(1/q)$ : about 19 at  $a = 0.9$  and  $2\xi$  near criticality, where  $\xi = 1/\log(1/a)$  is the correlation length. The loss is thus unbounded as correlations lengthen, so the full resolvent must be retained. Sharpness concerns truncation, not the full learned error or runtime. For fixed structural margins, bounded  $M_0$ , fixed  $\tau$  and local evaluation cost, the error floor  $O(\varepsilon + \sqrt{h})$  yields an online score-evaluation budget  $O(|I_r(S)|\delta^{-2}\log(1/\delta))$  for accuracy  $\delta$ , with  $h \asymp \delta^2$ , sufficient score accuracy and radius chosen from the spatial bound. Constants must include correlation length, graph growth and stability margins; training and certification are excluded. On  $\mathbb{Z}^D$ , where  $|I_r(S)|$  grows like  $r^D$ , a radius loss by a factor  $\varrho$  multiplies this budget by about  $\varrho^D$ .

### 3. Candidate and planned fellowship

**3.1 Competence, recruitment and independence.** We seek a recent PhD from outside Sweden in probability, stochastic analysis, optimal transport or numerical analysis, with evidence of independent mathematical work and the ability to lead a main result independently. Expertise in coupling or transport methods, Markov-chain perturbation theory or numerical analysis of stochastic differential equations is central. The position will be advertised internationally and filled in open competition. Friesen and I will lead mathematical assessment; Villanueva-Perez will assess the optional imaging connection. Katsoulakis is not on the selection committee.

Dr Hyemin Gu has expressed interest. She defended her PhD with Katsoulakis at UMass Amherst in September 2025 and is now a postdoc there. Her doctoral research concerns Wasserstein-regularised generative flows. Her first-author work on Lipschitz-regularised gradient flows and generative particle algorithms [23] supports the reference-law defect analysis; her residual-architecture work [24] brings experience converting growth and sensitivity bounds into stability certificates. Transport-based analysis and stability of learned dynamics are the two skills the project needs, and Friesen supplies complementary coupling expertise. At Lund she would develop a distinct line in spatial perturbation theory, within the programme I formulated, in a new group and with Friesen as a new mentor. The fellow will lead Theorem B, choose an extension and shape the research line at Lund, with contribution-based authorship and freedom to publish independently.

**3.2 Two-year programme (from 1 July 2027). Core success:** the combined A+B theorem for a certified nonconvex family with incompatible, own-coordinate-dependent residuals, explicit decay and a complete accuracy budget. The Gaussian benchmark and one controlled non-Gaussian test are essential; broader potentials and application studies are subsequent objectives. **Months 1–4:** establish the baseline and Gaussian checks, reproduce nonconvex estimates and begin global certification; benchmark metrics and identify the limiting constants. **Months 3–10:** prove spatial transfer and the finite-time recursion, including the convex-block case, for a first manuscript combining weighted boundary transfer, incompatible finite updates and an explicit computational guarantee. During the Amherst visit compare direct learned-kernel coupling with defect feedback; select the stronger certified regime by month 8. **Months 8–20:** prove Theorem B, transient moment bounds and global certification, and combine A+B for a second manuscript. Report admissible residual tolerances and deterioration near the stability boundary. At month 12, if core milestones permit, the fellow chooses between broader potentials and a fellow-led extension (signed analysis for frustrated Gaussian models, or typical-observation guarantees via disagreement percolation [25]), aiming at an independent result beyond the initial theorem. **Months 18–24:** compare certified and observed errors as radius, residual size and interaction strength vary, and release reproducible code. Graph-imputation work supplies testbed design; guarantees apply to the specified block sampler, not automatically to the original diffusion architecture. Sensor or MAX IV studies are optional validation, testing the consequences of violated locality. The fellow will lead a working seminar and a two-day national workshop, and prepare a Swedish Research Council starting-grant application, subject to eligibility. Articles will be openly accessible through Lund’s repository under KAW requirements.

**3.3 Risks and lasting value.** Weak contraction and excessive feedback may require improved envelopes or reflection metrics, while preserving the nonconvex goal. If the decay bound of Theorem A fails, we will identify the missing assumption through a counterexample and prove the corresponding conditional result. If feedback prevents retention of the reference exponent, establish the certified slower rate and identify the mechanism. Each milestone reports both admissibility and the full error multiplier. A convex fallback must still deliver spatially weighted reference-law guarantees for incompatible finite updates and artificial boundaries; a standard perturbation bound alone is insufficient. The lasting contribution is a Lund research line connecting probability and learned stochastic dynamics, with the fellow owning a main theorem and an independent extension. We will invite Jimmy Olsson (KTH), Fredrik Lindsten (Linköping) and Moritz Schauer (Chalmers) to the seminar and workshop, and seek participation in Mittag-Leffler’s *Frontiers in Optimal Control: Geometry, Complexity, and Learning* programme in autumn 2028.

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